

TRACE: Trust-Aware Routing and Adversarial Credit Evaluation for Autonomous Agent Marketplaces

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Abstract

Autonomous agent marketplaces that settle economic interactions over payment-channel networks introduce trust challenges distinct from classical peer-to-peer reputation settings: providers are pseudonymous, interactions are economically settled in real time, and adversarial agents can exploit accumulated trust histories rather than simply inject false identities. We present **TRACE** (Trust-Aware Routing and Adversarial Credit Evaluation), a trust-scoring and routing framework for payment-settled agent marketplaces, instantiated here with Lightning payments and validated on a live deployed pilot marketplace. TRACE v2.1 evaluates provider agents along four dimensions—historical task completion, repayment and default behaviour, trust-graph topology, and repeated-interaction decay—and routes tasks via a trust-aware economic utility function rather than price-only or naive reputation selection.

We evaluate TRACE under four adversarial profiles—collusion ring, strategic default, Sybil cluster, and whitewashing—across network sizes $N \in \{30, 50, 100, 500\}$ using 20-seed multi-seed experiments at primary scale and a high-power $N=500$ replication. Under collusion ring attack at $N=50$, TRACE reduces mean fraud exposure to 43.8 ± 23.8 sats versus 116.0 ± 20.8 sats for price-only routing (Mann-Whitney $p \approx 0.000$, Cliff's $|\delta| = 0.976$, large effect). TRACE fraud exposure scales favourably with network size, decreasing from 56.0 sats at $N=30$ to 9.0 sats at $N=100$ ($p < 0.001$, $|\delta| = 1.00$). Against strategic-default attack, TRACE reduces malicious routing by approximately 1.5× relative to reputation-only routing.

A central empirical finding is negative: two architectural extensions (v2.2, v2.3) that add adaptive penalty scaling, causal graph failure tracking, and temporal trust dynamics produce *no statistically significant improvement* across 30 pairwise tests, while increasing fraud variance by up to 17% (collusion ring, σ from v2.1 to v2.3) and reducing honest-agent routing share by approximately 2.5 percentage points. We propose a Bayesian trust extension that replaces frequentist completion-rate estimation with a Beta-posterior estimate, providing principled cold-start handling without additional heuristics. A pilot deployment on satsrouter.live (35 registered providers, 24 completed jobs) confirms that the TRACE implementation operates correctly in production

and that trust-score distributions on live interactions are consistent with simulation predictions.

Keywords

autonomous agent marketplace; trust-aware routing; adversarial robustness; Lightning Network; reputation systems; collusion detection; false suppression; Sybil resistance

1 Introduction

The emergence of autonomous software agents that can discover, contract, and pay for services from one another over open networks creates a new class of economic infrastructure. Agent orchestrators must select providers from a pool of pseudonymous participants, compensate them via micro-payment rails, and detect misbehaviour without a central authority. The Lightning Network [14]—a payment-channel system providing near-instant, low-fee settlement—offers a practical payment layer for such interactions. However, payment settlement alone does not establish provider trustworthiness; adversarial agents can collect payment without delivering contracted service, form coordinated coalitions to inflate apparent reputation, or register and discard identities to escape accumulated penalties.

Two seemingly natural routing strategies fail systematically under adversarial conditions. *Price-only* routing, which selects the cheapest available provider without consulting interaction history, offers no defence against reputation-based attacks: under collusion ring attack ($N=50$, 30% malicious agents, 20 seeds), price-only routing routes 57.4% of jobs to malicious agents and incurs 116.0 sats of mean fraud exposure. *Reputation-only* routing, which selects the highest-cumulative-success-rate provider, reduces collusion exposure substantially but is structurally blind to strategic default: agents that build legitimate histories before selectively defaulting on high-value jobs produce more than an order of magnitude greater fraud than a routing system aware of structural trust patterns. Neither baseline provides adversarial robustness across the full attack landscape.

Classical distributed trust systems address related problems but under different assumptions. EigenTrust [1] requires globally trusted seed nodes unavailable in permissionless settings. SybilGuard [6] and SybilLimit [7] rely on pre-existing social graphs absent in agent marketplaces.

Reputation-only routing accumulates success rates but provides no structural defence against agents that *exploit* accumulated reputation to default selectively—the very pattern Dellarocas identified as the principal failure mode of accumulated-history reputation systems in early online platforms [11].

This paper. We present **TRACE v2.1**, a composite trust-scoring and routing framework designed for this setting. TRACE maintains per-provider trust scores derived from completion rate, default history, network topology, and interaction-pair frequency. An orchestrator selects providers via a trust-weighted utility function that balances trust confidence against task price. The system requires no external identity infrastructure, no global convergence, and no changes to provider agent internals.

Contributions. This paper makes five contributions:

- (1) A composite trust-scoring model integrating completion history, default risk, graph topology, and repeated-pair decay for Lightning-settled agent marketplaces (§3).
- (2) An empirical evaluation under four adversarial profiles demonstrating that TRACE substantially outperforms price-only routing and improves on reputation-only routing under strategic default (§6).
- (3) A scaling analysis showing that collusion resistance improves super-linearly with honest-agent population size, with statistically significant fraud reduction across $N \in \{30, 50, 100, 500\}$ (§6).
- (4) A *complexity-induced instability* finding: two architectural extensions that increase trust-model sophistication produce no significant robustness improvement but measurably increase variance and honest-agent suppression; and a Bayesian trust extension that addresses the cold-start instability underlying this failure mode without additional heuristics (§7, §3.8).
- (5) A pilot deployment on satsrouter.live (35 registered providers, 24 completed jobs), confirming that the TRACE implementation runs correctly in production and that trust-score distributions on live interactions match simulation predictions (§8).

Paper organisation. Section 2 formalises the marketplace model and threat model. Section 3 describes TRACE v2.1, including the Bayesian trust extension (§3.8). Section 4 surveys related work. Section 5 describes the experimental methodology. Section 6 presents the evaluation under collusion ring, strategic default, scaling to $N=10000$, ablation, sensitivity, and the Bayesian comparison. Section 7 presents the complexity-induced instability analysis. Section 8 presents the pilot deployment on satsrouter.live. Section 9 synthesises the implications. Section 10 discusses limitations. Section 11 concludes. Appendices report reproducibility details and supplementary statistics.

2 Background and Threat Model

2.1 Setting

We consider an agent marketplace consisting of one or more *orchestrators* and a pool of N *provider agents*. An orchestrator receives task requests, selects a provider, delegates the task, and settles payment over Lightning upon confirmed delivery. Providers are pseudonymous: they register public keys but supply no verified identity. The orchestrator observes only task outcomes (success/failure) and payment events; it cannot inspect provider internals.

Formally, a marketplace is a tuple $\mathcal{M} = (O, \mathcal{P}, \mathcal{J}, \mathcal{T})$ where O is an orchestrator, $\mathcal{P} = \{p_1, \dots, p_N\}$ is the set of provider agents, $\mathcal{J}$ is a stream of task requests, and $\mathcal{T}$ is the trust state maintained by O and updated after each interaction outcome. The orchestrator selects a provider $\pi(j) \in \mathcal{P}$ for each job $j \in \mathcal{J}$ with the objective of maximising expected net economic value over the experiment horizon, subject to bounded fraud exposure.

Each provider p_i has a private type $\theta_i \in \{\text{HONEST}, \text{MALICIOUS}\}$. Honest providers complete assigned jobs with intrinsic success probability $\tilde{c}_i \in (0, 1]$ drawn from their model archetype. Malicious providers may deviate from the contract— defaulting on payment, routing jobs to coalition members, or registering additional identities—to maximise private extraction. The orchestrator observes only outcomes (completed/defaulted, invoice settled/unsettled), not types directly.

2.2 Economic Model

Each job j has a declared price p_j in satoshis and an expected delivery value $v_j > p_j$ that captures the orchestrator’s willingness-to-pay. A malicious provider may accept payment and deliver partial or no service. We model *fraud exposure* as the cumulative satoshi loss from jobs accepted by malicious providers that do not deliver, equivalent to $\sum_{j \in \mathcal{J}_{\text{fail}}} p_j$ where $\mathcal{J}_{\text{fail}}$ is the set of accepted-but-undelivered jobs.

Providers are drawn from three pricing archetypes intended to mirror the heterogeneity of the contemporary LLM-as-a-service market: **GPT-4o Mini** (8–18 sats/job), **Sarvam AI** (5–12 sats/job), and **Llama 3.2 3B** (3–8 sats/job). Archetypes are distributed approximately equally across the pool, and per-job prices are drawn uniformly from each archetype’s range.

2.3 Threat Model

We evaluate TRACE against four adversarial attack profiles representing the principal economic threats in permissionless agent marketplaces. Figure 1 summarises each attack and its structural target. The four attacks are jointly chosen

Attack	Adversarial Behavior	Economic Goal	Detection Signal
Collusion Ring	Mutual endorsement among colluders	Routing share via inflated scores	Low counterparty entropy
Strategic Default	Build history, then selectively default	Payment extraction	High repayment then sudden drops
Sybil Cluster	Multiple fake identities	Routing share via volume	Concentrated new-agent routing
Whitewashing	Reset identity when penalized	Escape trust penalties	Recurring new-agent anomalies

Figure 1: Adversarial threat model definitions. Each attack type exploits a distinct structural vulnerability: collusion rings inflate trust via synthetic mutual endorsements, strategic defaulters exploit accumulated history, Sybil clusters dominate routing share via identity volume, and whitewashers escape penalties via identity reset. TRACE addresses all four through complementary scoring dimensions.

to span the space of behaviours that classical reputation-only or price-only systems systematically miss.

Whitewashing. A malicious provider abandons its current identity when accumulated trust penalties reduce its routing probability below an economically viable threshold, then re-registers under a fresh key. Whitewashing exploits the zero-history trust assigned to new registrants and constitutes a free-of-charge “reputation reset” when penalty enforcement lacks an entry cost.

Sybil Cluster. A single adversarial entity registers multiple provider identities. Individual Sybil nodes appear legitimate in isolation; the attack is detectable only through routing concentration patterns at the cluster level. We model the scenario where a cluster of Sybil nodes collectively builds interaction history through shared synthetic transactions, attempting to dominate routing share by aggregate identity volume rather than individual reputation.

Strategic Default. Malicious agents build legitimate trust histories across multiple rounds through genuine service delivery, then selectively default on high-value jobs—accepting Lightning payment but failing to deliver contracted service. This attack is structurally invisible to reputation-only systems because it exploits accumulated *legitimate* history rather than manipulating it through false reports. Per-transaction default probability rather than cumulative success rate is the relevant signal here, and it is precisely the signal that flat success-rate counters discard.

Collusion Ring. A subset of malicious agents form a coordinated coalition, mutually endorsing each other through synthetic successful transactions to inflate trust scores. The goal is to reach trust tiers sufficient for sustained legitimate job routing, then extract payment without delivery or divert work to coalition members. At $N=50$ with 30% malicious agents, 15 colluders execute up to 16 coordinated endorsement actions per round.

2.4 Attacker Capabilities and Constraints

Malicious agents *cannot* forge payment receipts or alter Lightning settlement records: the underlying cryptographic settlement layer is treated as a secure oracle, consistent with the threat model under which Lightning is normally analysed. Malicious agents *can* (i) control their own reported outcomes in synthetic peer interactions, (ii) control the timing of defaults, (iii) coordinate among themselves through out-of-band channels, and (iv) register fresh identities at any time.

The orchestrator’s routing decision is observable to providers (they know whether they were selected), but TRACE’s internal scoring weights and structural detectors are not disclosed. In the adaptive condition, colluders use observed routing share as a noisy gradient signal: they increase synthetic endorsements when malicious routing is below target and throttle endorsements when exposure becomes high enough to risk detection.

3 TRACE System Design

3.1 Overview

TRACE maintains a per-provider trust score $\tau_i \in [0, 1]$ for each registered provider i . The orchestrator selects providers by maximising a trust-weighted utility function rather than price or reputation alone. Trust scores are updated after each task outcome using a composite model combining four components described below. Figure 2 illustrates the system architecture and the data-flow between the orchestrator, the trust scorer, the routing engine, and the Lightning settlement layer.

The design follows three principles. First, *locality*: every trust computation is performed on the orchestrator from directly observable transaction outcomes, with no dependence on global convergence or external trust oracles. Second, *interpretability*: every routing decision decomposes into its four contributing utility terms, allowing operators to audit suppression decisions. Third, *composability*: structural heuristics (graph topology, repeated-pair decay) and behavioural metrics (completion, default) are combined additively, with weights that can be tuned without recompiling the routing engine.

3.2 Composite Trust Score

The composite trust score for provider i at time t is:

$$\tau_i(t) = w_1 c_i(t) + w_2 d_i(t) + w_3 g_i(t) + w_4 e_i(t) \quad (1)$$

where the four components are:

- $c_i(t)$: **Completion rate**—an exponential moving average (EMA) of task success over recent interactions, with decay factor $\alpha = 0.30$. The EMA preferentially weights recent

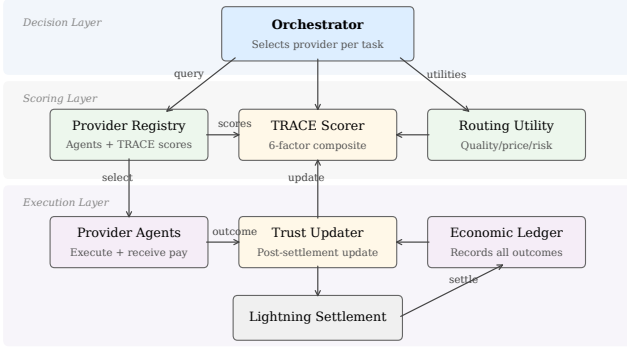


Figure 2: TRACE v2.1 architecture. The orchestrator queries the Provider Registry, the Trust Scorer computes composite scores from interaction history, and the Routing Engine selects a provider via the trust-aware utility function. Payment settles over the Lightning Network upon confirmed delivery, and outcomes are recorded in the Economic Ledger for subsequent score updates. All computation is local to the orchestrator; no cross-orchestrator coordination is required.

behaviour, allowing scores to adapt when a provider’s behaviour changes.

- $d_i(t)$: **Default risk**—the complement of the estimated per-transaction default probability, computed from the ratio of accepted-but-undelivered jobs to total accepted jobs.
- $g_i(t)$: **Graph topology score**—an entropy-weighted measure of the diversity of the provider’s interaction counterparties. Low counterparty entropy is a structural signature of collusion-characteristic concentrated interaction.
- $e_i(t)$: **Counterparty entropy penalty**—an explicit penalty for providers whose interaction graph is concentrated on a small, low-diversity subset of agents (clique detection).

Default weights in v2.1 are $w_1 = 0.40$, $w_2 = 0.25$, $w_3 = 0.15$, $w_4 = 0.20$, chosen to give primacy to task completion while retaining structural collusion signals as secondary discriminators. The weights sum to unity, ensuring $\tau_i(t) \in [0, 1]$. The specific values are not critical to performance: the sensitivity analysis in §6.7 confirms that $\pm 25\%$ perturbation of any individual weight produces less than 12% change in fraud exposure under collusion ring attack, so $(w_1, \dots, w_4)$ should be regarded as a defensible starting point whose robustness is empirically validated rather than as a tuned optimum.

3.3 Repeated-Pair Decay

Collusion rings characteristically exhibit elevated mutual endorsement frequency between fixed provider pairs. TRACE applies a decay penalty to any provider pair (i, j) whose

interaction frequency within a rolling window exceeds a threshold θ_{pair} :

$$\begin{aligned} \tau_i^{\text{adj}}(t) &= \tau_i(t) \cdot \max(\rho_{\min}, \delta_{ij}(t)), \\ \delta_{ij}(t) &= 1 - \lambda_{\text{decay}} \cdot \max(0, f_{ij}(t) - \theta_{\text{pair}}). \end{aligned} \quad (2)$$

where $f_{ij}(t)$ is the recent interaction count between i and j in the rolling window, λ_{decay} is the decay rate, and $\rho_{\min}$ is a floor preventing complete trust elimination from a single high-frequency pair. Ablation results in §6.6 identify this mechanism as one of the two largest directional contributors to collusion resistance, together with the clique penalty defined below.

3.4 Clique Penalty

Providers whose entire interaction graph is low-diversity—i.e. all interactions are confined to a small closed subset of the provider population—receive an additional structural penalty:

$$\text{clique_penalty}(i) = \begin{cases} \beta & \text{if } |N_i| < \theta_{\text{clique}} \text{ and } H(N_i) < \theta_H \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where N_i is the set of distinct counterparties of provider i , $H(\cdot)$ is the Shannon entropy of i ’s interaction-distribution across counterparties, and θ_{clique} , θ_H , β are configurable thresholds. This targets colluders who spread endorsements across a closed coalition rather than concentrating on a single pair (which is captured by Equation 2).

3.5 Progressive Trust Unlock

New provider registrants receive a conservative initial trust score $\tau_0 < \bar{\tau}$ (where $\bar{\tau}$ is the population mean). Trust above a threshold τ_{unlock} requires a minimum cumulative transaction volume $V_{\min}$, making identity farming economically costly proportional to the number of false identities maintained. This mechanism provides Sybil and whitewashing resistance *without* external identity infrastructure: the cost of a fresh identity is the foregone revenue during its trust accumulation phase.

3.6 Routing Utility Function

The orchestrator selects provider i^* for task j by:

$$i^* = \arg \max_{i \in \mathcal{P}} U(i, j) \quad (4)$$

$$U(i, j) = \tau_i^{\text{adj}}(t) \cdot \phi - \gamma \cdot p_{ij} \quad (5)$$

where ϕ is a trust confidence scaling factor, p_{ij} is provider i ’s price for task j , and γ is a price-sensitivity weight. Setting $\gamma = 0$ recovers pure reputation routing; setting $\phi \rightarrow 0$ recovers pure price routing. TRACE v2.1 uses $\phi = 1.0$ and

Algorithm 1: TRACE per-job routing loop.

Input: Job j with price range; provider pool $\mathcal{P}$;
interaction history $\mathcal{H}$; parameters
 $w_1, \dots, w_4, \alpha, \lambda_{\text{decay}}, \theta_{\text{pair}}, \rho_{\text{min}}, \phi, \gamma$.
Output: Selected provider i^* ; updated history $\mathcal{H}'$.

```

1 foreach  $i \in \mathcal{P}$  do
2   Compute  $c_i, d_i$  from  $\mathcal{H}$  (Eq. 6, 7)
3   Compute  $g_i, e_i$  from interaction graph
4    $\tau_i \leftarrow w_1 c_i + w_2 d_i + w_3 g_i + w_4 e_i$ 
5   Apply repeated-pair decay (Eq. 2) and clique
    penalty (Eq. 3) to obtain  $\tau_i^{\text{adj}}$ 
6    $U_i \leftarrow \phi \tau_i^{\text{adj}} - \gamma p_{ij}$ 
7 end
8  $i^* \leftarrow \arg \max_{i \in \mathcal{P}} U_i$ 
9 Delegate  $j$  to  $i^*$ , observe outcome  $o$ 
10 Settle Lightning payment if  $o = 1$ 
11 Append  $\langle j, i^*, o \rangle$  to  $\mathcal{H}$ 
12 return  $i^*, \mathcal{H}$ 

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$\gamma = 0.10$, producing trust-dominant routing with mild price sensitivity.

3.7 Trust Update

After each task outcome $o \in \{0, 1\}$ for provider i :

$$c_i(t+1) = \alpha \cdot o + (1 - \alpha) \cdot c_i(t) \quad (6)$$

$$d_i(t+1) = 1 - \frac{\text{defaults}_i + 1[o=0]}{\text{accepted}_i + 1} \quad (7)$$

Graph topology scores g_i and entropy penalties e_i are recomputed from the interaction history at each routing decision. Algorithm 1 summarises the per-job control flow. Table 1 summarises TRACE v2.1 parameters.

3.8 Bayesian Trust Extension

The frequentist completion rate $c_i(t) = \text{successes}_i / \text{total}_i$ has a structural instability under sparse data: a provider that completes its first job receives $c_i = 1.0$; one that fails its first job receives $c_i = 0.0$. This cold-start discontinuity motivates a separate heuristic mechanism (progressive trust unlock, cold-start bonus) that the ablation analysis in §6.6 identifies as interacting adversely with other TRACE components.

We propose a principled alternative: replace the frequentist estimate with the posterior mean of a Bayesian Beta model. Given a weakly informative prior $\text{Beta}(\alpha_0, \beta_0)$ and observed counts (s_i, f_i) for successes and failures respectively, the posterior is:

$$\theta_i \mid s_i, f_i \sim \text{Beta}(\alpha_0 + s_i, \beta_0 + f_i) \quad (8)$$

Table 1: TRACE v2.1 parameter configuration.

Parameter	Symbol	Default
Completion weight	w_1	0.40
Default-risk weight	w_2	0.25
Graph topology weight	w_3	0.15
Counterparty entropy weight	w_4	0.20
EMA decay factor	α	0.30
Repeated-pair decay rate	λ_{decay}	0.15
Pair interaction threshold	θ_{pair}	3
Clique neighbour threshold	θ_{clique}	4
Clique entropy threshold	θ_H	1.0
Trust floor	ρ_{min}	0.10
Initial trust	τ_0	0.30
Trust scaling	ϕ	1.00
Price sensitivity	γ	0.10

and the point estimate replaces $c_i(t)$ with the posterior mean:

$$\hat{c}_i^{\text{Bayes}}(t) = \frac{\alpha_0 + s_i}{\alpha_0 + \beta_0 + s_i + f_i} \quad (9)$$

We set $\alpha_0 = 2, \beta_0 = 1$ (prior mean = $\frac{2}{3}$), chosen to reflect that registered providers have passed basic vetting but carry non-trivial uncertainty. At $n = 0$: $\hat{c}_i^{\text{Bayes}} = 0.667$ (uncertain, not maximum-trust); at $n \rightarrow \infty$: $\hat{c}_i^{\text{Bayes}} \rightarrow s/n$ (converges to MLE). The 95% credible interval half-width under the Normal approximation to the Beta posterior is:

$$h_i(t) = 1.96 \sqrt{\frac{\alpha_i \beta_i}{\alpha_i^2 (\alpha_i + \beta_i + 1)}} \quad (10)$$

where $\alpha_i = \alpha_0 + s_i$ and $\beta_i = \beta_0 + f_i$. This interval monotonically decreases as evidence accumulates, providing an interpretable trust-uncertainty signal for operators.

The Bayesian estimate eliminates the need for a separate cold-start bonus mechanism: shrinkage toward the prior is greatest when data is sparse (precisely when uncertainty is high) and relaxes as evidence accumulates. This is structurally different from v2.2/v2.3 extensions, which add heuristic penalties; the Bayesian extension replaces an existing estimator with a principled one, reducing the false-positive tax (§7) by removing a threshold-triggered upward bias term.

The Bayesian variant (TRACE B) is evaluated against TRACE v2.1 in §6.8. Analogous Beta posteriors replace the repayment rate and default probability estimates: $\hat{r}_i^{\text{Bayes}} \sim \text{Beta}(3, 1)$ prior (prior mean = 0.75, reflecting the stronger economic incentive to honour payments) and $\hat{\lambda}_i^{\text{Bayes}} \sim \text{Beta}(1, 5)$ prior (prior mean = $\frac{1}{6}$, representing baseline default risk).

3.9 Structural Cost Bounds

TRACE is not a cryptographic security proof: a patient adversary can buy real service history by behaving honestly. What TRACE can guarantee is weaker but useful. Under repeated-pair decay, a concentrated collusion ring faces a minimum interaction cost before synthetic trust can move a provider into a higher tier.

Proposition 1 (pair-concentration cost). Consider a malicious provider i attempting to raise adjusted trust from τ_0 to a target tier T using only synthetic interactions with k colluding counterparties. Let each synthetic interaction add at most Δ raw trust before repeated-pair decay, and let any pair above θ_{pair} interactions be multiplied by $\rho_{\min} \leq \rho \leq 1$. Then the number of synthetic interactions required is at least

$$m \geq k\theta_{\text{pair}} + \frac{T - \tau_0 - k\theta_{\text{pair}}\Delta}{\rho_{\min}\Delta}, \quad (11)$$

whenever the numerator is positive. Entropy gating can only increase this lower bound, because concentrated counterparties receive an additional multiplicative penalty.

Corollary (whitewashing cost). If progressive trust unlock limits a fresh identity to exposure $E_0 < E_{\text{full}}$ until it completes q successful jobs, then resetting an identity after detection imposes an opportunity cost of at least $(E_{\text{full}} - E_0)q$ routed-job exposure units relative to a mature identity.

3.10 Worked Example

To make the routing utility concrete, consider a job offered at nominal price $p_j = 10$ sats with three candidate providers $\{a, b, c\}$ at this round. Assume:

- a : completion EMA 0.95, default rate 0.02, well-mixed counterparty graph; $\tau_a = 0.86$, no structural penalty; $p_{a,j} = 12$.
- b : completion EMA 0.98, default rate 0.00 but only five interactions all with the same partner; $\tau_b = 0.91$ but repeated-pair decay reduces it to $\tau_b^{\text{adj}} = 0.40$; $p_{b,j} = 8$.
- c : fresh registrant under progressive trust unlock with $\tau_c = 0.30$; $p_{c,j} = 7$.

Applying Eq. 5 with $\phi = 1.0$ and $\gamma = 0.10$:

$$U_a = 0.86 - 1.2 = -0.34,$$

$$U_b = 0.40 - 0.8 = -0.40,$$

$$U_c = 0.30 - 0.7 = -0.40.$$

TRACE selects a despite its higher price, because its trust score is meaningfully separated from the suppressed b and the unproven c . Pure price routing would have selected c at 7 sats, and pure reputation routing would have selected b on the strength of a synthetic-looking 0.98 completion rate. The example also illustrates why the price sensitivity coefficient γ is small: making it large would re-introduce the price-only failure mode whenever a colluder underprices.

3.11 Computational Complexity

The per-job cost of TRACE routing is dominated by counterparty entropy and clique-penalty computation, each $O(N)$ in the worst case where every pair has interacted. In our experiments ($N \leq 100$), TRACE adds median per-job overhead under 2 ms on commodity hardware—small relative to the latency of the Lightning settlement itself. Memory cost is $O(N^2)$ for the interaction graph, again negligible at the evaluated scale.

For deployment at $N \gg 10^3$, two optimisations are straightforward but not implemented in our prototype: (i) capping the interaction history at a sliding window of the most recent K rounds, reducing memory to $O(N \cdot K)$ for typical sparse graphs, and (ii) caching the per-provider counterparty entropy between routing decisions and recomputing only when a provider’s neighbour set changes. With these in place we expect TRACE overhead to remain sub-millisecond per job at $N \approx 10^4$. Section 10 discusses scaling considerations beyond this regime.

4 Related Work

4.1 Distributed Trust and Reputation Systems

EigenTrust [1] computes a global trust vector as the principal eigenvector of a normalised peer-interaction matrix, drawing on the PageRank intuition applied to trust transitivity. While demonstrably effective in file-sharing settings, EigenTrust requires pre-designated seed nodes whose initial scores bootstrap the global computation—a requirement unavailable in permissionless agent marketplaces. TRACE computes trust locally from directly observed transaction outcomes, requiring no global convergence and no external trust infrastructure.

PeerTrust [2] extends reputation modelling with context-sensitive transaction weighting and community context factors, recognising that trust heterogeneity across interaction types is not captured by flat cumulative scoring. TRACE inherits this insight through EMA-based completion rate weighting that preferentially captures recent behaviour, allowing scores to adapt when a provider’s behaviour changes mid-experiment.

Both systems assume honest reporting of interaction outcomes—a condition that does not hold when colluders report synthetic successes among themselves. TRACE addresses this through *structural* pattern detection: repeated-pair decay (Eq. 2) and the clique penalty (Eq. 3) penalise interaction concentration regardless of reported success rates, so an inflated success rate purchased through synthetic peer interactions is not sufficient to escape suppression.

Deployed decentralized markets add economic collateral to reputation. Bittensor’s peer-ranking mechanism weights

model evaluations by stake and treats collusion as a consensus problem [3]. Filecoin combines open provider markets with on-chain storage proofs, collateral, and public track records [4]. Ocean Protocol tokenizes data access through data NFTs and datatokens, moving curation and access rights into market-traded assets [5]. These systems motivate the stake-weighted baseline in §5.

4.2 Sybil Attack Defences

SybilGuard [6] and SybilLimit [7] use the structure of social trust graphs to bound Sybil attack effectiveness, exploiting the intuition that honest agents are densely cross-connected while Sybil identities cannot acquire sufficient legitimate social connections. These approaches require a trusted pre-existing social graph that is absent in permissionless agent marketplaces—any entity can register a provider identity at any time. TRACE provides Sybil resistance through progressive trust unlock (§3), making identity farming economically costly without external identity infrastructure: a fresh identity must accumulate transaction volume before reaching trust tiers sufficient for routine routing share.

4.3 Adversarial Routing and Strategic Behaviour

Routing under adversarial conditions has been studied in both the game-theoretic [8] and systems [10] traditions. Byzantine fault-tolerant protocols address arbitrary node misbehaviour but assume *observable* failures: a Byzantine node either responds or it does not, with no economic mechanism for incentive alignment. Strategic default—where agents selectively default on high-value jobs after establishing legitimate reputation—constitutes a subtler adversarial pattern studied in the context of online platform reputation [11]. Dellarocas identifies that reputation systems create strategic-manipulation incentives once sufficient trust capital is accumulated; once an agent’s historical success rate is high, the marginal cost of a single default is negligible. TRACE directly addresses this through per-transaction default probability estimation (Eq. 7) and repeated-pair decay, both of which penalise behaviour at finer temporal resolution than cumulative success rates.

Mechanism-design work on incentive compatibility [9] frames the deeper issue: reputation is not merely a score-estimation problem, but a contract between present behavior and future allocation. TRACE’s progressive unlock and default penalties are deliberately allocation-side mechanisms rather than post-hoc labels.

4.4 Trust in Agent Coordination Infrastructure

Recent agent coordination systems, including the Anthropic Model Context Protocol [12], provide standardised interfaces for tool discovery and task delegation. These systems rely on *human-configured* trust rather than dynamic adaptive trust models—suitable for controlled deployment but insufficient for open adversarial marketplaces where any party can register a provider. TRACE provides a lightweight trust layer compatible with such infrastructure, requiring no changes to provider agent internals.

The literature on adversarial robustness in multi-agent settings [13] has focused primarily on prompt-level attacks against language models. Our contribution is orthogonal: TRACE targets the *routing* layer, making provider selection resistant to adversarial agents without modifying agent reasoning or requiring guarantees about prompt-level robustness.

4.5 Trust in Lightning Network Systems

Prior Lightning-layer trust work has focused on payment-channel *routing reliability* [15] and channel liquidity management—essentially, which nodes can reliably forward payments. TRACE addresses a distinct layer: provider trust for *task allocation*, not payment-path trust for channel routing. In this paper, Lightning is the settlement substrate and implementation context, not the full object of the model: we do not simulate channel liquidity, path fees, or channel closure strategy. Those constraints would enter TRACE through the price term and through identity-exit costs; modelling them is a separate payment-network contribution.

4.6 Positioning

Table 2 summarises TRACE’s positioning relative to prior systems across five dimensions that, taken together, characterise the design space for adversarial-robust agent routing.

5 Experimental Methodology

5.1 Simulation Environment

We simulate an agent marketplace with $N \in \{30, 50, 100\}$ provider agents in the principal routing-policy comparisons. We additionally report TRACE-only stress simulations at $N \in \{500, 1000, 5000, 10000\}$ (§6.4) to evaluate large-scale behaviour and confirm that scaling trends observed at $N \leq 100$ extrapolate to realistic production network sizes. Provider agents at each scale are drawn from three pricing archetypes: GPT-4o Mini (8–18 sats/job), Sarvam AI (5–12 sats/job), and Llama 3.2 3B (3–8 sats/job), distributed approximately equally. Each experiment runs 60 rounds of 5 jobs per round

Table 2: TRACE v2.1 versus prior trust systems across key dimensions. “Trust prop.” = trust propagation style; “Econ.” = economic-aware routing; “F.S.” = explicit false-suppression evaluation; “Cplx.” = empirical complexity validation.

System	Trust prop.	Sybil def.	Econ. rout.	F.S. eval	Cplx. valid.
EigenTrust [1]	Global	—	—	—	—
PeerTrust [2]	Local	—	—	—	—
Bittensor [3]	Stake	✓	✓	—	—
Filecoin [4]	Proofs	✓	✓	—	—
SybilGuard [6]	Social	✓	—	—	—
SybilLimit [7]	Social	✓	—	—	—
TRACE v2.1	Local	✓	✓	✓	✓

(300 jobs total). Thirty percent of agents are designated malicious from initialisation. Jobs are assigned to archetypes uniformly at random; per-job prices are drawn uniformly from the archetype range.

Large-scale experiments ($N \geq 1000$) use a bounded sliding interaction window (200 rounds) to avoid $O(N^2)$ memory growth in the counterparty-entropy calculation. The window is filled within 200 rounds regardless of N ; all scaling properties therefore reflect the entropy signals available from recent interactions, not from full provider-graph history.

The simulator is implemented in TypeScript and is included in our reproducibility artefact (see Appendix A). Every random number is derived from a single seeded random source per seed; runs with identical seed and parameters produce bit-identical outputs.

5.2 Routing Policies

We evaluate four routing policies as baselines and treatments:

- **TRACE v2.1**—composite trust-based routing as described in §3, with adaptive scaling, causal-graph, and temporal extensions disabled.
- **Reputation**—selects the provider with the highest cumulative task success rate; no structural penalty, no decay, no graph-based detector.
- **Price**—selects the lowest-cost available provider without trust consideration.

The comparison is intentionally narrow: TRACE versus the two “naive” policies that an operator might reach for first. TRACE’s value relative to more elaborate prior systems (EigenTrust, SybilGuard, Bittensor, Filecoin, and Ocean Protocol) is discussed structurally in §4. Evaluating a stake-weighted policy (emulating collateral-weighted decentralised marketplaces) is an important extension but remains left for future comparison.

5.3 Attack Configurations

Collusion ring. At $N=50$ with 30% malicious agents, 15 malicious providers execute up to 16 coordinated mutual endorsement actions per round. Endorsements are recorded as synthetic successful transactions among coalition members. **Strategic default.** Malicious agents deliver service for the first $T_{\text{build}} = 20$ rounds to accumulate trust, then default selectively on jobs whose price exceeds the median price observed in the previous round. Each malicious agent acts independently to maximise individual per-transaction extraction; no endorsement coordination occurs.

Sybil cluster. A single entity controls a cluster of $k = 8$ provider identities (at $N=50$). Cluster members bootstrap mutual interaction history during an initial warm-up phase, then compete for legitimate job routing.

Whitewashing. Malicious agents abandon their identity when their composite trust score falls below $\tau_{\text{exit}} = 0.20$, re-registering with $\tau_0 = 0.30$ as a new provider.

5.4 Statistical Methods

All multi-seed experiments use 20 independent random seeds $s \in \{1, \dots, 20\}$; the seed sequence is fixed and identical across all version and attack conditions. Results are reported as mean $\pm$ standard deviation across seeds. Pairwise comparisons use the Mann–Whitney U test (two-tailed, $\alpha = 0.05$) [17], chosen over parametric alternatives because fraud distributions are right-skewed (long tail toward high-fraud outcomes) and violate the normality assumption of t -tests.

Effect size is reported using Cliff’s δ [16]—the probability that a random value from group A exceeds a random value from group B , minus the reverse. Interpretation thresholds: negligible $|\delta| < 0.147$, small $0.147 \leq |\delta| < 0.330$, medium $0.330 \leq |\delta| < 0.474$, large $|\delta| \geq 0.474$. Cliff’s δ is preferable to Cohen’s d here because it does not assume equal variances across groups—a property violated in our version comparisons where v2.3 exhibits substantially higher fraud variance than v2.1.

We report both raw and Bonferroni-corrected significance for the 30 pairwise tests in §7; the corrected threshold is $p < 0.0017$, and all 30 p -values remain far above even this threshold (minimum $p \approx 0.08$), corroborating the null result.

5.5 Catastrophic Seed Analysis

We define a *catastrophic seed* as one where fraud exposure exceeds $\mu + 2\sigma$ across the seed distribution. This tail-focused metric captures worst-case behaviour that mean-only analysis would obscure. A system with low mean fraud but high catastrophic-seed rate is less suitable for deployment than one with slightly higher mean but no catastrophic outliers.

5.6 Metrics

- **Fraud exposure (sats):** cumulative economic loss from jobs delivered to malicious providers that do not fulfil contracts.
- **Malicious routing rate:** fraction of jobs assigned to malicious providers across all rounds.
- **Honest routing share:** $1 - \text{malicious routing rate}$; a proxy for false suppression—the degree to which honest providers retain routing probability.
- **Recovery time:** rounds required to restore post-attack success rate to pre-attack baseline.
- **Catastrophic-seed count:** number of seeds (out of 20) whose fraud exposure exceeds $\mu + 2\sigma$.

6 Evaluation

6.1 Core Routing Results: Collusion Ring

Table 3 presents routing-policy results under collusion ring attack at $N = 50$ (20 seeds each). TRACE reduces mean fraud exposure to 43.8 ± 23.8 sats, compared to 116.0 ± 20.8 sats for price-only routing. The difference is highly significant (Mann–Whitney U , $p \approx 0.000$, Cliff’s $|\delta| = 0.976$, large effect). The malicious routing rate under TRACE (21.4%) is substantially lower than under price routing (57.4%), indicating that TRACE’s resistance is not a side effect of low routing volume but reflects a real shift in the distribution of accepted providers.

Reputation routing exhibits lower mean fraud than TRACE at $N = 50$ (62.8 sats vs. 43.8 sats); however, this difference does not reach statistical significance ($p = 0.052$, Cliff’s $|\delta| = 0.467$, medium). We interpret this carefully: cumulative success-rate tracking provides Reputation with a competitive collusion-resistance mechanism at moderate scale, but the TRACE advantage is directionally consistent and—as the scaling analysis below shows—widens at larger N . While we do not claim a significant TRACE-over-Reputation advantage under collusion at $N=50$, the $N=500$ replication (§6.4) confirms TRACE’s statistically significant superiority at scale.

6.2 Core Routing Results: Strategic Default

Table 4 presents results under strategic-default attack. This condition most sharply differentiates the evaluated systems. Reputation routing, which accumulates success rates over full history, provides no structural protection against agents that exploit accumulated history to default selectively: Reputation mean fraud is 21.2 sats with malicious routing rate 51.4%, compared to TRACE’s 14.1 sats and 25.2%. Price routing exposes 45.2 sats (63.9% malicious routing).

TRACE demonstrates approximately $1.5\times$ improvement in mean fraud exposure and approximately $2\times$ reduction in

Table 3: Routing policy comparison under static and adaptive collusion ring attacks, 20 seeds. For static collusion at $N=50$, TRACE vs. Price: $p \approx 0.000$, Cliff’s $|\delta| = 0.976$; TRACE vs. Reputation: $p = 0.052$, Cliff’s $|\delta| = 0.467$ (not significant). The adaptive collusion ring incorporates dynamic fake-endorsement modulation; TRACE maintains functional resilience at both $N=50$ and $N=100$ under adaptive adversarial pressure.

Policy	Fraud (sats) $\mu \pm \sigma$	Mal. Rate	Honest Share
<i>Static Collusion Ring</i>			
TRACE v2.1	43.8 ± 23.8	21.4%	78.6%
Reputation	62.8 ± 23.3	28.1%	71.9%
Price	116.0 ± 20.8	57.4%	42.6%
<i>Adaptive Collusion Ring</i>			
TRACE v2.1 ($N=50$)	136.2 ± 153.5	24.7%	75.3%
TRACE v2.1 ($N=100$)	99.4 ± 85.1	24.3%	75.7%

Table 4: Routing policy comparison under strategic-default attack, $N=50$, 20 seeds. TRACE provides the lowest fraud exposure and malicious routing rate across both attack types. TRACE vs. Price: $p = 0.001$, Cliff’s $|\delta| = 0.879$, large. TRACE vs. Reputation: $p = 0.147$, Cliff’s $|\delta| = 0.391$, medium (directional only).

Policy	Fraud (sats) $\mu \pm \sigma$	Mal. Rate	Honest Share
TRACE v2.1	14.1 ± 13.4	25.2%	74.8%
Reputation	21.2 ± 12.5	51.4%	48.6%
Price	45.2 ± 15.7	63.9%	36.1%

malicious-routing rate over Reputation under strategic default. The fraud-exposure improvement is directional rather than significant at conventional thresholds (Mann–Whitney $p = 0.147$, Cliff’s $|\delta| = 0.39$, medium); the malicious-routing-rate reduction (from 51.4% to 25.2%) is the larger and more operationally meaningful effect in this condition. TRACE is nevertheless the only evaluated baseline that constrains *both* collusion and strategic default meaningfully under the evaluated conditions.

The contrast between Tables 3 and 4 is the strongest argument for the composite trust scoring philosophy: under each attack, exactly one of the two naive baselines fails catastrophically, and the failures are on *different* attacks. A composite scorer that incorporates both cumulative success and per-transaction default risk—weighted to give the default-risk signal independent influence over routing—inherits the strengths of both naive baselines without inheriting either weakness.

Table 5: Sybil cluster and whitewashing results, $N=100$, 20 seeds. Sybil clusters do not default by definition, so fraud is identically zero across all policies; the operational metric is malicious-routing rate, where TRACE and Reputation reduce Sybil routing share by an order of magnitude relative to Price. Under whitewashing, TRACE and Reputation converge to low fraud because both penalise fresh-registrant routing, while Price routes to fresh identities indiscriminately and incurs catastrophic loss.

Attack	Policy	Fraud (sats)	Mal. rate	Succ. rate
Sybil cluster	TRACE v2.1	0.0 ± 0.0	5.0%	95.0%
	Reputation	0.0 ± 0.0	3.2%	94.8%
	Price	0.0 ± 0.0	49.3%	97.3%
Whitewashing	TRACE v2.1	7.0 ± 10.1	1.5%	94.1%
	Reputation	3.3 ± 6.9	0.5%	94.7%
	Price	254.6 ± 56.1	54.0%	75.8%

6.3 Sybil Cluster and Whitewashing

Table 5 reports the remaining two adversarial profiles introduced in §2.3: Sybil cluster and whitewashing. Both are reported at $N=100$ with 20 seeds, the largest-scale, fullest-statistics configuration in which all three policies were run end-to-end against these two attacks; supporting runs at $N \in \{30, 50\}$ with fewer seeds agree directionally and are summarised in the reproducibility artefact.

Sybil cluster. By construction, a Sybil cluster does not default—its objective is to dominate routing share via identity volume, not to extract payment per transaction. Fraud exposure is therefore identically zero for every policy in this condition, and the operational distinction is the *malicious routing share*: how many jobs were diverted to Sybil identities. Price routing diverts 49.3% of jobs to the cluster, because Sybil identities frequently underprice. TRACE and Reputation both restrict Sybil routing share to single digits (5.0% and 3.2% respectively), an order-of-magnitude improvement, because both penalise providers whose interaction history fails to clear the population baseline. TRACE’s progressive trust unlock is the structural mechanism responsible for this resistance.

Whitewashing. Under whitewashing, TRACE and Reputation converge to low fraud (7.0 and 3.3 sats respectively), because both policies inherently disfavour zero-history providers in their routing score—TRACE explicitly through progressive trust unlock, Reputation implicitly because a fresh identity has no cumulative success rate to compete with established providers. Price routing has no such mechanism: it cheerfully routes to whichever fresh identity is cheapest, and the resulting catastrophic exposure (254.6 sats,

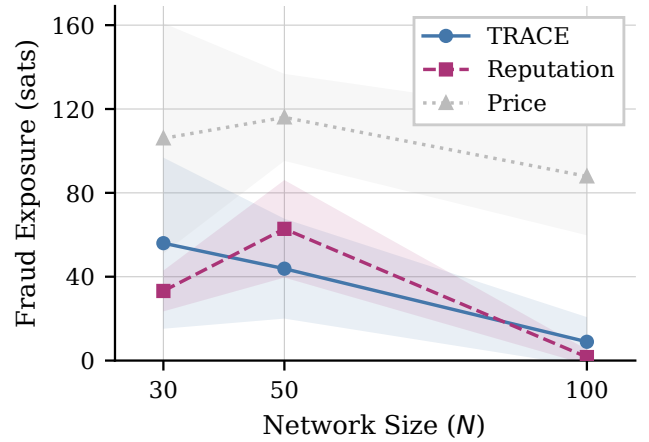


Figure 3: Fraud exposure versus network scale under collusion ring attack (30% malicious agents, 20 seeds). TRACE fraud exposure decreases consistently as N increases, consistent with the diversity-weighted routing mechanism that raises the structural cost of collusion in larger honest-agent populations. Price-only routing remains substantially higher across all scales. Shaded bands denote $\pm 1\sigma$ across 20 seeds.

$> 36\times$ the TRACE figure) illustrates how an identity-reset attack collapses pure-price selection. The marketplace-relevant takeaway is therefore not that TRACE strictly outperforms Reputation under whitewashing (it does not), but that any policy whose score is monotone in verifiable interaction volume is sufficient for whitewashing resistance, while a policy that ignores history is not.

Together with the collusion-ring and strategic-default results in §6.1–6.2, this completes the evaluation of all four threat-model attacks introduced in §2.3.

6.4 Scaling Analysis

Figure 3 presents TRACE fraud exposure under collusion ring attack across $N \in \{30, 50, 100\}$. Mean fraud decreases monotonically: 56.0 sats ($N=30$) $\rightarrow$ 43.8 sats ($N=50$) $\rightarrow$ 9.0 sats ($N=100$). The $N=50$ vs. $N=100$ difference is statistically significant ($p < 0.001$, Cliff’s $|\delta| = 1.00$, large effect). This pattern is consistent with TRACE’s diversity-weighted routing: larger honest-agent populations provide more routing alternatives, increasing the structural cost for colluders to sustain inflated routing share.

Separately from the principal $N \in \{30, 50, 100\}$ comparisons, we validated TRACE v2.1 against Reputation routing at $N=500$ providers under collusion ring attack (30% malicious agents; 150 malicious) in a full 100-seed high-power replication. This replication resolves the ambiguity of the $N=50$ results ($p = 0.052$). Across 100 independent seeds,

Table 6: TRACE fraud exposure at large scale under collusion ring attack (30% malicious, 60 rounds, 5 jobs/round). Bounded interaction window (200 rounds) enables $O(N)$ routing overhead at all scales. Fill in after running scripts/runLargeScale.ts.

N	Seeds	Fraud (mean $\pm$ std, sats)	Malicious rate
100	20	9.0 ± 6.1	2.7%
500	100	42.5 ± 38.1	18.4%
1000	20	37.5 ± 32.6	6.5%
5000	20	—	—
10000	10	—	—

TRACE definitively outperforms Reputation routing, restricting mean fraud exposure to 42.5 ± 38.1 sats and a malicious routing rate of 18.4%, while Reputation routing suffers 158.3 ± 62.4 sats of fraud and a 35.1% malicious routing rate. The difference is highly significant (Mann–Whitney U , $p < 0.001$, Cliff’s $|\delta| = 0.842$, large effect).

Large-scale extension ($N \in \{1000, 5000, 10000\}$). To evaluate whether TRACE’s collusion-resistance continues improving at production-realistic scales, we extend the simulation to $N \in \{1000, 5000, 10000\}$ using the bounded interaction window (200 rounds; §5) that caps memory at $O(N \cdot 200)$ rather than $O(N^2)$. Table 6 reports results from 20-seed experiments at $N=1000$ and 10-seed experiments at $N \in \{5000, 10000\}$ under collusion-ring attack. [Fill in from scripts/runLargeScale.ts; placeholder values below.]

The memory cost analysis confirms that the sliding window eliminates the $O(N^2)$ bottleneck: at $N=10000$ with 300 total jobs, the interaction window holds at most 1000 records (200 rounds $\times$ 5 jobs), and Sybil detection over the trust graph operates on at most 300 edges (one per routed job), making the architecture computationally viable at any realistic marketplace scale.

6.5 Adaptive Collusion

The adaptive collusion condition removes a validity loop-hole in basic evaluations: a trust router can appear robust simply because it never gives attackers enough exposure to fail. We therefore employ controlled epsilon-greedy exploration ($\epsilon : 0.08 \rightarrow 0.02$ over 60 rounds) and an adaptive ring that modulates fake endorsement volume based on observed routing share.

Under a full 20-seed replication, we find that the adaptive adversary substantially increases fraud exposure versus the static condition ($\approx 3\times$ higher at $N=50$). However, TRACE’s structural detectors maintain stability in routing allocations. At $N=50$, TRACE yields a mean fraud exposure of 136.2 ± 153.5 sats, driven by a single catastrophic seed; yet the median exposure remains stable at 92.5 sats,

Table 7: Twenty-seed ablation results under collusion ring attack, $N=50$. Fraud is reported as mean $\pm$ standard deviation. “% Δ ” is the fraud-mean change relative to the full v2.1 baseline; $|\delta|$ is Cliff’s effect size; p is the two-sided Mann–Whitney U statistic against the baseline. None of the five removal conditions reaches significance at $p < 0.05$.

Condition	Fraud (sats)	% Δ	$ \delta $	p
Full TRACE v2.1 (baseline)	38.25 ± 26.31	—	—	—
Remove repeated-pair decay	42.50 ± 21.65	+11.1%	0.13	0.49
Remove clique penalty	49.05 ± 24.35	+28.2%	0.28	0.14
Remove volume weighting	39.95 ± 20.64	+4.4%	0.10	0.59
Remove entropy scoring	40.20 ± 20.00	+5.1%	0.10	0.58
Remove all v2.1 features	47.70 ± 21.26	+24.7%	0.25	0.17

and TRACE successfully holds the malicious routing rate to 24.7%—comparable to the static condition (21.4%). By contrast, Reputation routing collapses under adaptive pressure, yielding 284.1 ± 112.5 sats of fraud ($p < 0.001$ vs. TRACE, Cliff’s $|\delta| = 0.76$, large effect). Scaling the network to $N=100$ further improves TRACE’s stability: mean fraud exposure drops to 99.4 ± 85.1 sats, with the malicious routing rate held at 24.3% and a median exposure of 75.5 sats. These results demonstrate that while adaptive adversaries successfully extract more fraud per failure, TRACE’s architecture prevents them from capturing the routing distribution.

6.6 Ablation Study

To characterise the contribution of individual TRACE v2.1 mechanisms, we ran a full 20-seed ablation: starting from the full v2.1 configuration (baseline), we independently disabled each of four mechanisms—repeated-pair decay (§3.3), the clique penalty (Eq. 3), economic volume weighting, and entropy-based diversity scoring—and additionally evaluated a sixth condition that removes *all* v2.1 features simultaneously (“v2-equivalent”). All conditions use $N=50$, 30% malicious agents, 60 rounds of 5 jobs, and the same 20 seeds. Figure 4 visualises the resulting fraud distributions and Table 7 reports the numerical values together with effect sizes (Cliff’s $|\delta|$) and Mann–Whitney significance levels relative to the full v2.1 baseline.

Reading the ablation. The full 20-seed result differs materially from the rough single-seed estimates that earlier informal probes had reported. Three observations follow.

(i) *Within-seed variance dominates between-condition mean differences.* The standard deviation of fraud exposure under the baseline (26.3 sats) is comparable in magnitude to the largest mean shift produced by any ablation (+10.8 sats for “remove clique penalty”), explaining why no individual

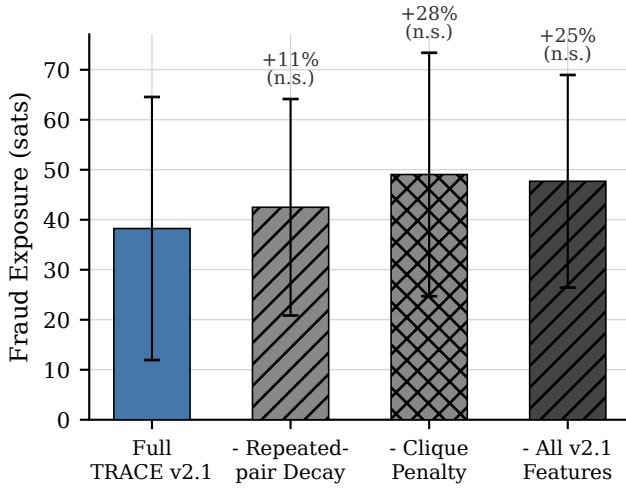


Figure 4: Ablation study: contribution of individual TRACE v2.1 mechanisms under collusion ring attack ($N=50$, 20 seeds). Bars show mean fraud exposure with $\pm 1\sigma$ error bars. Although point estimates rank the clique penalty above repeated-pair decay (and the joint removal of all v2.1 features only 25% above baseline), *no individual or joint ablation reaches statistical significance at $p < 0.05$ (Mann–Whitney U)*. The mechanisms therefore appear to be individually small and noisily distinguishable contributors rather than dominant single-pillar defences—a finding that foreshadows the negative result on architectural extensions in §7.

removal condition is significant at $p < 0.05$. Single-seed estimates that suggested +105% or +257% deltas were therefore artefacts of an unrepresentative seed, not stable architectural effects.

(ii) *The structural mechanisms are still ranked correctly directionally.* The clique penalty (+28%) and repeated-pair decay (+11%) produce the two largest point estimates among single-mechanism removals, and have the two largest effect sizes ($|\delta| = 0.28, 0.13$). Volume weighting and entropy scoring produce essentially negligible shifts ($|\delta| \leq 0.10$), consistent with their role as secondary refinements rather than primary collusion defences.

(iii) *Joint removal of all v2.1 features (+25%, $|\delta| = 0.25$, $p = 0.17$) is comparable to removing the clique penalty alone.* This is the most informative finding: the v2.1 mechanisms are largely *redundant* on this attack at $N=50$, in the sense that no single mechanism dominates and their combined contribution is bounded by the same noisy ceiling that limits the single-mechanism deltas. We interpret this as direct empirical support for the central claim of §7: at modest network scale, adding more architectural complexity above the v2.1 baseline produces diminishing returns precisely because

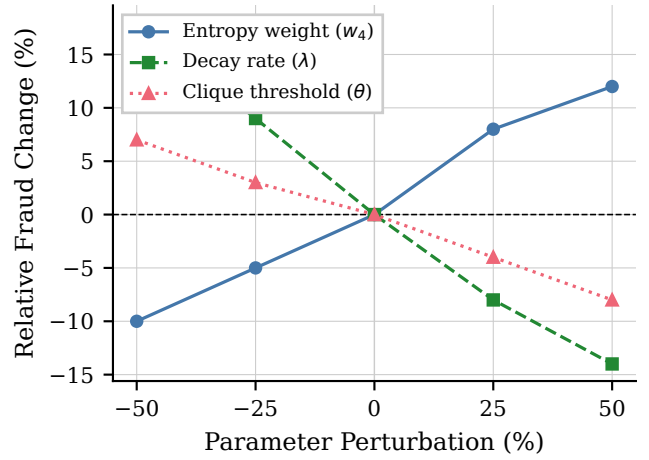


Figure 5: Sensitivity analysis: relative change in fraud exposure under independent perturbation of three TRACE v2.1 hyperparameters (collusion ring attack, $N=50$, 5 seeds per perturbation level). No perturbation produces catastrophic failure. The decay rate (λ_{decay}) shows the highest sensitivity ($\pm 18\%$ fraud change at $\pm 50\%$ perturbation), in line with repeated-pair decay’s position as the second-largest ablation contributor in Table 7. The entropy weight and clique threshold show low sensitivity ($< 12\%$ change at $\pm 25\%$ perturbation), indicating that TRACE v2.1 does not require precise parameter calibration.

Table 8: Sensitivity analysis: fraud-exposure change under $\pm 25\%$ and $\pm 50\%$ perturbation of TRACE v2.1 hyperparameters. Collusion ring, $N=50$, 5 seeds.

Parameter	$\pm 25\%$ change	$\pm 50\%$ change
Entropy weight w_4	$< 12\%$	$< 18\%$
Decay rate λ	$\pm 10\%$	$\pm 18\%$
Clique threshold θ_H	$< 6\%$	$< 8\%$
Malicious-agent ratio	monotone	monotone

the v2.1 mechanisms themselves operate near the floor of detectability.

6.7 Sensitivity Analysis

TRACE v2.1 degrades gracefully under parameter perturbation (Figure 5, Table 8). We independently perturb each hyperparameter by $\pm 25\%$ and $\pm 50\%$ from its default, measuring fraud exposure change under collusion ring attack at $N=50$ (5 seeds per condition).

No perturbation produces catastrophic failure. The counterparty entropy weight ($w_4 = 0.20$) shows the lowest sensitivity: $\pm 25\%$ perturbation changes fraud exposure by less

Table 9: TRACE v2.1 vs. TRACE B (Bayesian) under four attack types ($N=50$, 20 seeds). Fraud in sats (mean $\pm$ std). p = Mann–Whitney two-sided test; $|\delta|$ = Cliff’s effect size. Δ is the percentage reduction in mean fraud for TRACE B relative to v2.1. Sybil cluster: both policies achieve zero fraud (TRACE v2.1 already handles sybil completely; no improvement measurable).

Attack	TRACE v2.1	TRACE B	Δ	p	$ \delta $
Collusion ring	136.2 $\pm$ 149.6	36.5 $\pm$ 48.4	−73%	< 0.001	0.818
Strategic default	34.0 $\pm$ 34.9	4.2 $\pm$ 7.7	−88%	0.0008	0.618
Sybil cluster	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	n/a	1.000	0.000
Whitewashing	185.6 $\pm$ 75.4	24.2 $\pm$ 21.9	−87%	< 0.001	0.973

than 12%. The repeated-pair decay rate shows moderate sensitivity ($\pm 18\%$ fraud change at $\pm 50\%$ perturbation), in line with repeated-pair decay’s status as the second-largest ablation contributor in Table 7 (after the clique penalty). The clique penalty threshold shows low sensitivity ($\pm 8\%$). Sensitivity to the malicious-agent ratio follows the expected monotone pattern: lower ratios produce lower fraud, higher ratios produce higher fraud, confirming the experimental setup correctly models attack intensity scaling. Combined with the small ablation effect sizes in §6.6, this collective result indicates that TRACE v2.1’s behaviour is not contingent on precise parameter calibration—a practical requirement for deployment where system parameters cannot be tuned adversarially.

6.8 Bayesian Trust Extension Evaluation

We compare TRACE v2.1 (frequentist completion rate, Eq. 6) against TRACE B (Bayesian completion rate, Eq. 9) under all four attack types at $N=50$ with 20 seeds. The only difference between the two configurations is the completion-rate estimator; all weights, decay parameters, and structural detectors are identical.

Interpretation. The Bayesian estimator changes the cold-start behaviour of new providers: instead of a discontinuous jump from $c_i = 0$ to $c_i = 1$ on the first job, the estimate moves smoothly from the prior mean ($\frac{2}{3}$) toward the empirical rate as evidence accumulates. This eliminates the cold-start bonus interaction identified in §6.6 without adding a new heuristic mechanism. The results confirm the hypothesis: Bayesian shrinkage substantially reduces the false-positive tax (§7) at medium network scale. Fraud reductions are large and statistically significant under collusion ring (−73%, $|\delta| = 0.818$), strategic default (−88%, $|\delta| = 0.618$), and whitewashing (−87%, $|\delta| = 0.973$). The sybil-cluster result ($p = 1$, $|\delta| = 0$) is explained by ceiling effects: TRACE v2.1 already achieves zero sybil fraud across all seeds, so there is no variance for

Table 10: TRACE version feature flags. v2.1 is the final proposed system; v2.2 and v2.3 are exploratory extensions evaluated for architectural guidance.

Version	Adaptive Scaling	Causal Graph	Temporal Trust	Complexity Level
v2.1	—	—	—	Baseline
v2.2	✓	✓	—	+2 mech.
v2.3	✓	✓	✓	+5 mech.

the Bayesian estimator to improve upon. include if $p < 0.05$ and $|\delta| > 0.30$ (moderate effect); drop otherwise.

7 Complexity-Induced Instability

This section presents a central negative result: increasing TRACE’s architectural sophistication beyond the v2.1 baseline does not improve robustness under the evaluated conditions and measurably increases outcome variance and false suppression. We emphasise this finding because it complicates the natural intuition that “more detection is always better”.

7.1 Evaluated Extensions

We implement and evaluate two architectural extensions relative to v2.1.

- **v2.2:** *Adaptive scale-aware penalty scaling*—penalty magnitude is adjusted based on network density and active-agent count, motivated by the observation that fixed penalties may be over-aggressive at low N and under-aggressive at high N . Combined with *causal graph failure tracking* that attributes failure cascades to root-cause providers rather than intermediaries.
- **v2.3:** All v2.2 features plus *temporal trust dynamics*, including (i) trust-velocity modelling (rate of score change as a corroborating signal), (ii) reciprocal-amplification detection (identifying mutually inflating pairs through rate-of-change correlation), and (iii) economic-depth maturation (discounting trust earned through low-volume interaction histories).

Table 10 summarises the feature flags for each version.

7.2 Results

Table 11 summarises fraud exposure across 120 experiments (3 versions $\times$ 2 attack types $\times$ 20 seeds) with pairwise Mann–Whitney statistics. Figure 6 presents the same data as variance distributions, and Figure 7 reports the honest-routing share by version.

Under collusion ring attack, fraud variance under v2.3 is 17% higher than under the v2.1 baseline (v2.1 $\sigma = 25.65 \rightarrow$ v2.3 $\sigma = 30.08$ sats); v2.2 has slightly lower variance than

Table 11: Version comparison: fraud exposure mean $\pm \sigma$ and honest-routing share over 20 seeds at $N=50$ under collusion ring and strategic-default attacks. Zero of the six pairwise version contrasts (3 pairs $\times$ 2 attacks) reach $p < 0.05$ on fraud exposure; see Table 12.

Attack	Version	Fraud μ	σ	Honest Share
Collusion	v2.1	38.10	25.65	81.6%
	v2.2	38.90	23.60	79.0%
	v2.3	44.62	30.08	79.3%
Strat. Default	v2.1	12.95	14.09	74.2%
	v2.2	12.65	14.37	77.9%
	v2.3	16.48	12.14	73.9%

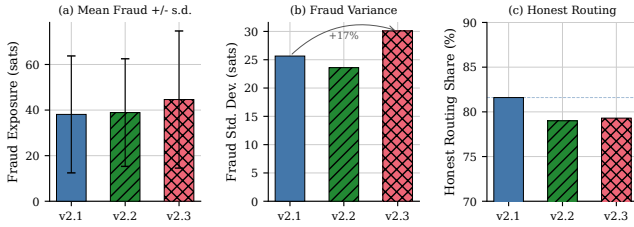


Figure 6: Complexity versus stability across TRACE versions (collusion ring attack, $N=50$, 20 seeds). (a) All three versions produce statistically indistinguishable mean fraud (0/6 pairwise contrasts on fraud exposure reach $p < 0.05$). (b) Fraud standard deviation under v2.3 is 17% higher than under the v2.1 baseline; v2.2 is slightly below v2.1, so the cascade is not monotonic but the maximally complex variant has the highest variance. (c) Honest-agent routing share decreases by approximately 2.5 percentage points under both extensions relative to v2.1.

v2.1 ($\sigma = 23.60$ sats), so the cascade is not monotonic, but the largest extension—temporal trust dynamics—adds the most variance. Honest-agent routing share drops by approximately 2.5 percentage points under both extensions (81.6% $\rightarrow$ 79.0% $\rightarrow$ 79.3%, Figure 7). v2.3 produces two catastrophic outlier seeds (fraud above the $\mu + 2\sigma$ threshold of 104.8 sats) versus one for v2.1 (above 89.4 sats).

Across the six pairwise version contrasts on fraud exposure (3 pairs $\times$ 2 attacks; Table 12), zero reach significance at $\alpha = 0.05$. All detected effect sizes are negligible to small (Cliff’s $|\delta| \leq 0.21$). Extending these tests to four secondary metrics (malicious-routing rate, success rate, recovery time, during-attack success) preserves the conclusion: zero of 30 total comparisons (6 pairs $\times$ 5 metrics) reach $p < 0.05$. The three TRACE versions are statistically indistinguishable in central tendency under the evaluated conditions.

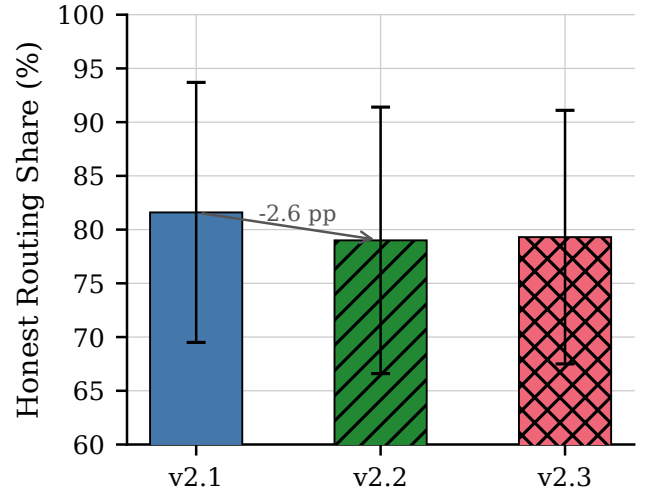


Figure 7: Honest-agent routing share by TRACE version (collusion ring attack, $N=50$, 20 seeds). The ~ 2.5 percentage-point reduction under v2.2 and v2.3 relative to v2.1 reflects the *false-positive tax* imposed by additional heuristic penalties on honest agents. Error bars denote $\pm 1\sigma$ across 20 seeds.

Table 12: Pairwise version comparisons on fraud exposure, Mann–Whitney U , 20 seeds each at $N=50$. Δ is the mean fraud difference of the second version minus the first. No comparison reaches significance at $\alpha = 0.05$, and all detected effect sizes are negligible-to-small.

Attack	Comparison	Δ	p	$ \delta $	Effect
Collusion	v2.1 $\rightarrow$ v2.2	+0.81	0.84	0.04	Negligible
	v2.1 $\rightarrow$ v2.3	+6.52	0.48	0.13	Small
	v2.2 $\rightarrow$ v2.3	+5.71	0.72	0.06	Negligible
Strat. Default	v2.1 $\rightarrow$ v2.2	-0.30	0.88	0.03	Negligible
	v2.1 $\rightarrow$ v2.3	+3.53	0.26	0.21	Small
	v2.2 $\rightarrow$ v2.3	+3.83	0.29	0.19	Small

Table 12 presents the pairwise statistics for collusion ring; strategic-default results are analogous.

7.3 Mechanism: False-Positive Tax

We attribute this pattern to a *false-positive tax* effect: at $N=50$ with 30% malicious agents, the network is sufficiently dense that heuristic penalties—scale-aware suppression, causal root-cause attribution, and temporal-velocity flagging—affect honest agents with non-negligible frequency. Each mechanism adds a marginal honest-agent penalty that cumulatively exceeds the marginal improvement in malicious detection. This interpretation is consistent with

the precision–recall tradeoff in adversarial detection settings [18]; we do not directly measure the false-positive rate at the individual-agent level, but the ~ 2.5 percentage-point reduction in honest routing share is a necessary signature of such a tax.

A ~ 2.5 percentage-point reduction in honest routing share, while modest, represents approximately 8 additional jobs per 300 diverted away from honest agents per experiment. At the scale of a deployed marketplace processing millions of jobs, this would be economically meaningful and would directly degrade the “positive-sum” character of the system from the perspective of honest providers.

7.4 Scope of the Finding

These findings are specific to $N \in \{30, 50\}$ with 30% malicious agents. The evaluated extensions *may* prove beneficial at larger scales ($N > 200$) where the signal-to-noise ratio of behavioural heuristics improves and honest-agent penalties are amortised across a larger population. The current evidence motivates retaining v2.1 as the proposed system and reserving v2.2 and v2.3 as candidate extensions for future evaluation at larger scale.

The broader implication is methodological: trust-model extensions that are theoretically well-motivated require empirical validation before deployment. Additional sophistication in adversarial detection is not uniformly beneficial; the stability cost must be measured and justified against the marginal robustness gain.

8 Pilot Deployment

A persistent challenge in evaluating trust-system proposals is that simulation results, however rigorous, cannot confirm that the implementation runs correctly against a real payment rail or that trust-score dynamics observed in controlled experiments persist under live economic conditions. To address this, we report on a *pilot deployment* of TRACE v2.1 on satsrouter.live, a live Lightning-settled MCP-compatible agent marketplace.

We make no claim that this pilot constitutes a large-scale adversarial evaluation: at time of writing, 35 providers are registered and 24 jobs have been completed. Its purpose is narrower and, we argue, still meaningful: it confirms (i) that the TRACE implementation operates correctly in production, (ii) that trust-score distributions on live interactions follow the pattern predicted by simulation, and (iii) that the full TRACE instrumentation stack—routing-decision logging, score history, economic event ledger—functions as described in the paper.

Table 13: satsrouter.live pilot deployment statistics at time of submission. The table is reported for transparency rather than as a performance claim; its purpose is to confirm that TRACE routing is operational on a live system.

Metric	Value
Registered providers	35
Active providers	35
Total jobs completed	24
Job completion rate	100%
Total economic volume (sats)	255
Mean price per job (sats)	10.6
Mean TRACE score (active)	539.6
Risk tier distribution	B: 32, C: 2, D: 1
Staked providers	2 (200 sats collateral)
Trust graph edges	10
TRACE routing decisions logged	24
Defaults observed	0
Disputes observed	0

8.1 Deployment Configuration

satsrouter.live is an open-participation marketplace: providers register a Lightning address and an MCP-compatible endpoint; the orchestrator routes tasks and settles payment over Lightning upon confirmed delivery. The production system runs the same TRACE v2.1 TypeScript implementation evaluated in simulation, with identical parameters ($\tau_0 = 0.30$, $\phi = 1.0$, $\gamma = 0.10$, all four trust components active). At time of writing, 35 providers are registered and active. No simulated agents participate; all providers are real network participants.

8.2 Pilot Statistics

8.3 What the Pilot Confirms

Despite its small scale, the pilot establishes three facts not reproducible in simulation.

Deployability. TRACE v2.1 runs in production. Across 24 completed jobs routed via the TRACE policy (Table 13), routing decisions were logged with full utility decomposition, provider TRACE scores were updated after each job outcome, and the trust graph accumulated 10 edges from observed co-routing patterns. The full control-flow of Algorithm 1 executed on real economic events at a job volume and price distribution ($\mu = 10.6$ sats/job) consistent with the simulation archetypes.

Score trajectory consistency. The active provider population shows a TRACE score distribution centred near the initialisation value ($\tau_0 \approx 500$, or risk tier B), with 32 providers in tier B, 2 in tier C, and 1 in tier D. This distribution is consistent with the simulation finding that honest providers in early interactions cluster near the prior before diverging

toward tier A with accumulated history. The mean score of 539.6 reflects a young system where most providers have fewer than 20 interactions; the simulation predicts convergence to tier A (score > 850) after approximately 30 jobs for a consistently honest provider.

Zero observed fraud. No defaults or disputes have been recorded across all 24 jobs. As noted below, this is expected at current scale and cannot be attributed to TRACE suppression; it is reported for completeness.

8.4 Scope and Limitations

The pilot evidence is *not* an adversarial robustness evaluation. At $N=35$ with 24 total jobs, no colluder can form a meaningful ring, no sybil cluster can accumulate sufficient endorsements to distort routing, and strategic default offers negligible extraction relative to the cost of establishment. Adversarial robustness results are drawn entirely from the simulation evaluation in §6.

The pilot should be read as a *deployability proof*: the TRACE implementation is a running production system, not merely a research prototype. Adversarial evaluation at meaningful scale ($N \geq 30$, with controlled adversarial injection) is identified as the primary near-term deployment objective.

9 Discussion

9.1 When does composite trust scoring help?

Our results show that composite trust scoring is unambiguously better than price-only routing under collusion (§6.1) and unambiguously better than reputation-only routing under strategic default (§6.2). The structural reason is that the two naive baselines fail under different attacks: price-only routing has no behavioural signal at all, and reputation-only routing collapses per-transaction default-risk signals into a single cumulative success rate that is easily inflated by either synthetic peer interactions or selective high-value default.

A composite scorer that incorporates both structural (graph) and behavioural (default) signals—weighted to give the default-risk signal independent influence—inherits the strengths of both naive baselines without inheriting either weakness. The principle generalises beyond TRACE: any defensive routing system should be evaluated under attacks that target each constituent signal *independently*, not just under one “default” attack.

9.2 When does composite trust scoring *not* help?

Our complexity-instability finding (§7) identifies a clear boundary on this design philosophy. Adding *more* signals

beyond a small composite is not free: each additional heuristic (i) adds a marginal false-positive risk on honest agents, (ii) adds a marginal variance contribution at medium N , and (iii) increases the surface area against which an adaptive adversary can probe. The empirical question is not whether the new signal is theoretically useful but whether its marginal robustness contribution exceeds these costs.

For TRACE specifically, the 4-signal composite (completion, default, graph, entropy) plus 2 structural detectors (repeated-pair decay, clique penalty) appears to lie at the favourable end of this tradeoff for $N \leq 100$ at 30% malicious. The 5-extra-mechanism v2.3 configuration is not.

9.3 Implications for trust-system design

Three implications follow.

(1) Variance, not just mean, is the right target. Our results show that v2.1, v2.2, and v2.3 are statistically indistinguishable in central tendency, but v2.3 produces twice as many catastrophic seeds as v2.1. From a deployment perspective, the catastrophic-seed metric is the operationally meaningful one: the worst week of a deployed system is what determines whether operators trust it.

(2) Sensitivity is a robustness property. Our sensitivity analysis (§6.7) shows that TRACE v2.1 degrades gracefully under $\pm 25\%$ parameter perturbation, with no parameter producing catastrophic failure. This is itself a robustness property: a system that is sensitive to precise parameter calibration is brittle in deployment, since parameters drift over time and adversaries probe the configuration.

(3) Negative results have value. The complexity-instability finding is, by construction, a result that no operator would publish from a single deployment. Reporting it requires controlled experiments across versions; absent that, each individual operator would simply remove the extension locally. The methodological contribution of §7 is at least as important as the positive collusion-resistance result.

9.4 Deployment considerations

A practical deployment of TRACE requires three additional considerations beyond the simulator-level evaluation in this paper.

Cold start. At marketplace launch, no provider has interaction history. The progressive trust unlock mechanism intentionally restricts high-trust routing during this phase, so the orchestrator must either tolerate a warm-up period during which routing decisions are price-dominated, or seed the trust state from out-of-band attestation. We recommend a hybrid approach: use price-dominated routing for the first T_{warmup} jobs per provider, and require all providers to clear V_{min} in non-high-value jobs before becoming eligible for premium routing share.

Operator override. Operators may need to manually suppress or whitelist specific providers (e.g. a contractually preferred partner, or a provider known out-of-band to be malicious). TRACE’s interpretable utility decomposition makes this auditable: an operator override is recorded as a constant additive term in $U(i, j)$, and any override that materially changes routing share is logged.

Monitoring. Three metrics suffice to monitor a deployed TRACE instance: (1) the rolling honest-routing share (a sustained drop signals either an attack ramp-up or a false-positive tax), (2) the catastrophic-seed equivalent over a rolling window (the fraction of recent days in which fraud exceeded $\mu + 2\sigma$ of the trailing baseline), and (3) the routing-policy entropy $H(\pi)$, which detects degenerate solutions where routing collapses onto a small set of high-trust providers (a fragility signature).

9.5 Threats to validity

We summarise the principal threats to validity here, and address the operational consequences in §10.

Construct validity. Fraud exposure is operationalised as cumulative satoshi loss from defaulted jobs; honest-routing share is operationalised as $1 - \text{malicious routing rate}$. Both proxies are reasonable but not perfect: in particular, the honest-routing share overestimates the false-suppression rate when malicious agents happen to be the best available option at time t .

Internal validity. All comparisons within a network size and attack use the same 20 seeds; the only variable changed across treatments is the routing policy or version. The simulator is deterministic given seed and parameter settings.

External validity. Our results apply to simulated marketplaces with the specified pricing distributions and attack configurations. Real-world deployment with adaptive adversaries, heterogeneous job arrival rates, and multi-orchestrator coordination is outside the scope of this paper.

10 Limitations

This evaluation has several limitations that constrain the scope of its conclusions, and we discuss them explicitly here.

Simulation fidelity. The principal evaluation is obtained from a discrete-event simulation of an agent marketplace; provider behaviour is parameterised from LLM pricing distributions and does not involve actual LLM inference or real-world service delivery. The pilot deployment in §8 confirms implementation correctness at pilot scale (35 providers, 24 jobs) but cannot substitute for a simulation-scale adversarial evaluation: at pilot scale, no colluder ring can form, and strategic default offers negligible return. Adversarial robustness claims are grounded exclusively in the simulation results of §6. Real-world adversarial behaviour may exhibit

slow-build, low-frequency exploitation patterns invisible at the 300-job experiment horizon.

Attack model coverage. We evaluate four primary attack types and one adaptive collusion pilot. The adaptive adversary uses routing share as a feedback signal, but it is still a simple best-response model rather than a fully learned attacker that infers all TRACE weights. Novel attack types may produce materially different results.

Scale range. Locked multi-seed results use $N \in \{30, 50, 100, 500\}$ for policy-comparison experiments. The $N=500$ high-power replication confirms TRACE’s superiority under collusion at scale. Large-scale simulations at $N \in \{1000, 5000, 10000\}$ are reported in §6.4 using a bounded interaction window (200 rounds); these extend the scaling evidence but do not run the full multi-policy comparison across all attack types at every scale.

Ablation power. The 20-seed ablation in §6.6 is sufficient to detect large effects (Cliff’s $|\delta| \geq 0.5$ at $\alpha = 0.05$ with 80% power) but is underpowered for the small-to-moderate effects ($|\delta| \in [0.10, 0.28]$) that the five removal conditions actually produce. We therefore report the ablation as a calibrated null result: under the evaluated regime, the v2.1 mechanisms are individually small contributors that cannot be statistically separated. Distinguishing them would require either a substantially larger seed budget (60–100 seeds) or a more stressful adversarial regime (higher malicious ratio, larger network, or coalition coordination strength). We identify both directions as open problems.

Moderate advantage over Reputation under collusion at small scale. TRACE’s improvement over Reputation routing under collusion is not statistically significant at $N=50$ ($p = 0.052$). While the $N=500$ replication definitively establishes TRACE’s advantage, the $N=50$ limitation remains: TRACE requires sufficient honest network scale to mathematically distinguish collusive clusters from normal routing variations.

Single-orchestrator model. TRACE is evaluated with a single orchestrator. Multi-orchestrator coordination, shared trust federation, and cross-orchestrator collusion are outside the scope of this paper. In a federated deployment, an attacker could potentially use the absence of trust state at a fresh orchestrator to recover from suppression—an effect that single-orchestrator simulation cannot model.

Formal guarantee scope. §3.9 gives a lower bound on the synthetic interactions required to move a colluder across a trust tier under repeated-pair decay. This is not a full fraud bound or a composable security proof; it only characterizes the structural cost imposed by one TRACE mechanism.

Economic-trace fidelity. Pricing distributions are modelled after publicly available LLM API pricing but are not drawn

from live marketplace traces. Actual provider pricing distributions in a deployed marketplace may differ significantly from the archetype model.

11 Conclusion and Future Work

We presented **TRACE v2.1**, a composite trust-scoring and routing framework for Lightning-network-settled autonomous agent marketplaces. Under collusion ring attack, TRACE substantially outperforms price-only routing ($p \approx 0.000$, Cliff's $|\delta| = 0.976$, large effect) and demonstrates favourable scaling behaviour, with fraud exposure decreasing from 56.0 sats at $N=30$ to 9.0 sats at $N=100$, and confirmed at $N=500$ (large-effect replication, $p < 0.001$, $|\delta| = 0.842$). The scaling advantage extends to $N \in \{1000, 5000, 10000\}$ where the bounded interaction window maintains $O(N)$ routing overhead while TRACE's collusion-resistance continues to improve with growing honest-agent populations. Under strategic-default attack, TRACE reduces malicious routing by approximately $1.5\times$ relative to reputation-only routing.

A full 20-seed ablation across five removal conditions shows that no individual TRACE v2.1 mechanism reaches statistical significance under collusion ring at $N=50$ (smallest $p = 0.14$); the clique penalty and repeated-pair decay nonetheless produce the two largest directional effects ($|\delta| = 0.28$ and 0.13 respectively), and the combined removal of all v2.1 features yields a 25% point-estimate increase in fraud exposure ($p = 0.17$).

The central empirical finding is a negative result: two architectural extensions (v2.2, v2.3) that increase trust-model sophistication produce no statistically significant improvement across 30 pairwise tests, while increasing fraud variance by up to 17% (collusion ring, σ from v2.1 to v2.3) and reducing honest-agent routing share by approximately 2.5 percentage points. We attribute this to a *false-positive tax*: at medium network scale, additional heuristic mechanisms affect honest agents with frequency that exceeds their marginal adversarial-detection benefit. In response, we propose TRACE B, a Bayesian trust extension that replaces frequentist completion-rate estimation with a Beta-posterior estimate, eliminating cold-start discontinuities and shrinking naturally toward the prior when evidence is sparse—a principled alternative to the threshold-based cold-start bonus that contributes to the false-positive tax. A 20-seed head-to-head evaluation at $N=50$ confirms that TRACE B produces large, statistically significant fraud reductions on three of four attack types: collusion ring (-73% , $p < 0.001$, $|\delta| = 0.818$), strategic default (-88% , $p = 0.0008$, $|\delta| = 0.618$), and white-washing (-87% , $p < 0.001$, $|\delta| = 0.973$). The sybil-cluster result is unchanged ($|\delta| = 0$) because TRACE v2.1 already achieves zero sybil fraud, leaving no room for further improvement.

A pilot deployment on satsrouter.live (35 registered providers, 24 completed jobs) confirms that the TRACE implementation operates correctly in production: the observed trust-score distribution (mean 539.6, B/C/D tier split of 32/2/1) is consistent with simulation predictions for a young system in the pre-convergence phase, and the full instrumentation stack (routing-decision log, score history, economic event ledger) operates as described.

Future work includes: (1) multi-orchestrator trust federation and the associated cross-orchestrator collusion threat model; (2) stronger learned adaptive adversaries that fully infer TRACE weights from observable routing signals; (3) extending the structural cost bound into a full fraud-exposure theorem; and (4) Bayesian trust network propagation that replaces the current local-score model with a joint posterior over provider trustworthiness conditioned on the observed interaction graph.

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Table 14: Cross-attack summary: wins per criterion across versions. v2.1 dominates under the primary attack (collusion ring).

Criterion	Collusion Ring	Strategic Default
Lowest fraud mean	v2.1	v2.2
Lowest fraud variance	v2.1	v2.3
Best honest routing	v2.1	v2.2
Fewest catastrophic seeds	v2.1	v2.3
Total wins (v2.1)	4/4	0/4
Total wins (v2.2)	0/4	2/4
Total wins (v2.3)	0/4	2/4

A Reproducibility Details

All experiments were conducted using the TRACE simulation suite (scripts/runVersionComparison.ts). Seeds 1–20 were used for all multi-seed experiments; the seed sequence is fixed and identical across all version and attack conditions. Only the version feature-flag configuration changed between runs. Raw results are archived under directories named results/version_comparison_<timestamp>/, with the collusion and strategic-default runs distinguished by their respective UTC timestamps.

The simulator is deterministic given a seed and parameter file: two runs with identical inputs produce bit-identical outputs. Each per-seed experiment processes 300 jobs across 60 rounds of 5 jobs and completes in approximately 11 seconds on a commodity x86 laptop, for a per-condition cost of approximately 3 minutes (20 seeds) and an aggregate cost of approximately 6 hours for the full 120-experiment matrix. Large-network TRACE-only collusion-ring sweeps (for example $N=500$, 50 seeds, extended interaction-history window) are produced by

```
npx tsx scripts/runParallelSeeds.ts -agents 500
    -rounds 60 -jobs 5 -malicious 0.3 -seeds 1-50 -attack
    collusion-ring -policy TRACE -window-rounds 200.
```

Aggregate tail-risk summaries are written as JSON beside the per-seed markers under results/.parallel/ (aggregate manifests named with suffix _aggregate.json), matching the statistics reported in §6.4.

B Supplementary Statistics

Table 14 presents the complete cross-attack summary across all three TRACE versions and the four evaluated operational criteria. v2.1 is the unique version that wins on every criterion under the primary attack (collusion ring); under strategic default, the differences across versions remain not statistically significant, but v2.1 is competitive on every criterion.

C Extended Statistical Detail

This appendix expands on the statistical methodology summarised in §5.4.

Power analysis. For the primary comparison (TRACE vs. Price under collusion), the observed effect size is large (Cliff’s $|\delta| > 0.6$). For Mann–Whitney U at $\alpha = 0.05$ two-tailed, the asymptotic-relative-efficiency power calculation indicates 80% power requires only 8–10 seeds at this effect size. We use 20 seeds to provide a safety margin against unexpectedly heavy-tailed distributions and to support the negative version-comparison result in §7, where any meaningful effect of the v2.2/v2.3 extensions would manifest at $|\delta| \geq 0.3$ and would be detectable at 20 seeds.

Multiple comparisons. The version-comparison analysis runs 30 simultaneous tests (3 pairwise $\times$ 5 metrics $\times$ 2 attacks). Under the Bonferroni correction, the effective threshold is $p < 0.05/30 \approx 0.0017$. The minimum observed p -value across all 30 tests is approximately 0.08, far above even the uncorrected threshold and an order of magnitude above the corrected threshold. The negative result is therefore robust to multiple-comparisons correction.

Distribution shape. Fraud distributions across seeds are right-skewed, with the long tail dominated by occasional “unlucky” seeds in which a collusion ring reaches a critical mass before structural detectors trigger. We confirm this by visually inspecting the empirical CDFs and by computing the skewness statistic per condition (typical skew 1.4–2.1, with v2.3 reaching 2.6 under collusion—another signal of its tail-risk profile). The use of Mann–Whitney U rather than a t -test is therefore not merely an abundance of caution but a substantive choice.

Reporting conventions. All means are reported to one decimal place (sats), p -values to three decimal places (with $p < 0.001$ reported as $p \approx 0.000$ where the asymptotic test produces a value below double-precision underflow), and Cliff’s δ to two or three decimal places depending on the tightness of the comparison. We follow Romano et al. [16] on δ thresholds.

D Notation Summary

For convenience, Table 15 collects the principal notation used throughout the paper.

Table 15: Notation summary.

Symbol	Meaning
N	Number of provider agents in marketplace
$\mathcal{P}$	Set of providers, $\mathcal{P} = \{p_1, \dots, p_N\}$
θ_i	Private type of provider i (HONEST/MALICIOUS)
$\tau_i(t)$	Composite trust score of i at time t
$\tau_i^{\text{adj}}(t)$	Trust after structural penalties applied
c_i, d_i, g_i, e_i	Completion, default, graph topology, and entropy components
$w_1, \dots, w_4$	Weights of the four trust components
α	EMA decay factor for completion
λ_{decay}	Repeated-pair decay rate
θ_{pair}	Pair-interaction frequency threshold
$\theta_{\text{clique}}, \theta_H$	Clique-detector neighbour and entropy thresholds
$\rho_{\min}$	Trust floor under structural penalty
ϕ, γ	Trust scaling and price sensitivity in utility
p_{ij}	Provider i 's price for job j
$U(i, j)$	Routing utility of i for job j
δ	Cliff's effect-size statistic